

POWER



Power is the speedy application of force. Dave Graham powering through *Jaws II*, 5.15a, Rumney.

Lee Hansche

CHAPTER

7

"To give the maximum effort at the precise moment it [is] needed. This is power." — Fred Nicole²

Muscular power is a universal requirement in athletics and a fundamental discriminator in many sports. Generally speaking, *power* is the ability to generate a large amount of force in a short period of time. Jumping, sprinting, and explosive lifts are good examples of activities that require power. This area of athletic performance has been studied extensively to improve performance in Olympic and ball sports, and much of that knowledge can be applied to climbing. Climbers define power as the ability to quickly exert high force on a set of holds. *Climbing power* is typically needed to execute cruxy, dynamic moves between small holds. This ability is essential to all free climbers. Even on a 30-pitch big wall, success or failure can come down to a single powerful move. No aspect of climbing performance presents such a clear-cut limiting factor as a climber's power. As the great Tony Yaniro says: "If you cannot pull a single move, you have nothing to endure."⁹

Improving one's power is critical to long-term improvement. As a climber rises through the grades, the crux moves become more powerful, and though progress can be made initially with improvements to other performance aspects, eventually, he will face moves that are too powerful. The specter of getting shut down by one or two "stopper" moves on a given goal route is always waiting in the future, at a higher grade. It only takes a single, too-hard move to doom a project. In order to have a fighting chance, the climber must be able to do the moves, and the hardest moves require power.

MAXIMUM RECRUITMENT, POWER, AND CONTACT STRENGTH

Maximum recruitment, power, and contact strength are terms used often by climbers in training, but their actual meanings and interrelationships can be somewhat ambiguous. When skeletal muscle performs a contraction, not all motor units are contributing to that contraction in a useful way. Some motor units may contract at the wrong time, or too slowly to be effective, and some may not contract at all. Some of this is accidental, but some is intentional. Individual motor units can only sustain a contraction for a fraction of a second, so sustained muscular contractions are achieved by alternating the brief contractions of individual motor units (allowing some to rest while others work).¹¹ This is an elegant trade-off for meeting the various demands placed on the muscles, but it necessitates careful training to align the competing traits of endurance and maximum force production with an athlete's goals. Fortunately, understanding how human skeletal muscle adapts to external stimulation makes it possible to design effective training for all styles of climbing.

MAXIMUM RECRUITMENT

The term *recruitment* describes the proportion of active motor units contributing to a given muscular contraction. By recruiting more units, greater force can be produced. Maximum Recruitment Training (aka Max R) seeks to condition the muscle to increase the proportion of motor units activated for a given contraction. In addition, Max R training improves force production on a second front: by improving the force production of each individual motor unit. The *size*



Contact Strength: if you can't generate high force quickly, you won't be able to latch dynamic moves. Brad Gobright sticking to *The Backbone*, 5.13a, Smith Rock, OR.  Brian Mosbaugh

principle states that smaller, slow-twitch motor units tend to be activated first, and the larger and stronger fast-twitch motor units tend to be activated last. If a given contraction does not require high force production, the large motor units are never activated. Lower-intensity strength training may fail to activate the largest units. High-intensity strength training, such as Max R training, demands activation of the strongest motor units, stimulating them to adapt.¹¹ However, Max Recruitment is only one-half of the power equation (the force element). The other half is the speed element.

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What separates *power* from *strength* is the speed or quickness of the contraction. The importance of speed in a muscular contraction is apparent to jumpers, sprinters, and other ball-sport athletes. The very hardest climbing moves usually have dynamic components as well, demanding rapid and precisely timed application of force. This is where speed becomes vital to the performance-oriented rock climber. Power is essential to both ends of any dynamic movement. First, power is needed to initiate the movement, creating upward momentum. Although usually (with good technique) the bulk of upward momentum is generated in the lower body, the fingers and upper body will experience a sudden, speed-dependent increase in load as the body moves upward, peaking when one hand leaves the rock to lunge for the next hold. In the latter half of a dynamic movement, the climber latches a target handhold, sometimes while “falling away.” This requires the forearm muscles to rapidly apply a great amount of force (*power!*).

CONTACT STRENGTH

The ability to rapidly latch a hold depends on *contact strength*, a mysterious term that is often misunderstood. During any contraction, muscles do not exert peak force instantaneously; rather, it can take up to a few tenths of a second to reach peak force. This is easy to demonstrate and worth experimenting with. Once properly warmed up, grab

a flat edge and slowly lift your feet off the ground, allowing the muscles time to produce maximum force. Next grab the same hold and quickly pull your feet off the ground. Notice how much more difficult it is to generate force the second time, instantly (and hang from the hold). Contact strength is the amount of force the fingers can generate during the period of initial contact with a given hold. This is critical during dynamic movements, because the fingers need to generate sufficient force as quickly as possible, during the brief instant when they first contact the target hold. If the dyno has been executed perfectly, the hold can be latched during the *deadpoint* (the point of time when the body is at the apex of flight, moving neither up nor down). Catching the hold at the deadpoint requires much less force. In most cases the climber will be falling back down onto the hold, requiring exponentially more force as he falls away. The sooner force can be generated, the less force is required.

MUSCLE FIBER TYPES

Clearly the ability to rapidly apply force is essential, as it is a fundamental component of contact strength, but how can it be improved? As discussed in **Chapter 5: Base-Fitness**, each muscle is composed of many motor units, each consisting of many individual muscle fibers. Currently there are seven known muscle-fiber types in humans, grouped into three categories. As you'll recall from that chapter, each fiber type behaves slightly differently. Type I fibers contract slowly, but are able to sustain longer duration activities. Type IIa and Type IIb (aka Type IIx/d) fibers are unable to sustain long contractions, but tend to be larger (and therefore stronger), and contract much more quickly than Type I fibers.

Fortunately for athletes it is possible to convert some fibers from one type to another, or to train fibers to behave like other fibers (by increasing mitochondria, glycolytic enzymes, fiber diameter, or contraction speed), so there is significant potential for climbers to train their muscles to behave in a way that suits their individual goals, regardless of genetics.^{10, 11}

PLYOMETRIC TRAINING

To succeed on powerful, dynamic moves, climbers must be able to apply a great amount of muscular force in a short amount of time. This can be done by:

- Increasing muscle-fiber recruitment
- Cultivating a higher distribution of fast-twitch fibers
- Training slow-twitch fibers to contract more quickly

Fortunately, there are specific training activities that accomplish all of these. Max-recruitment training in strength athletes is well understood, and it would be relatively simple to transfer this approach to climbing by using the typical finger-training tools, with adjusted repetitions and sets to achieve the desired muscular adaptations (this approach is described later in the chapter under “Max-Recruitment Training with a Hangboard”). Unfortunately, the tools used for developing static finger strength do little to improve contraction speed; enter *plyometrics*.

Plyometric training was originally developed by Soviet track and field coaches in the 1960s to help train explosive power in their athletes. Early plyometrics involved activities such as jumping from an elevated box, down to the ground, then immediately springing back up onto the box. Theoretically, the landing causes an involuntary eccentric contraction in the leg muscles, which must be immediately converted to a concentric contraction in a very short period of time (ideally less than 0.2 seconds to achieve a *true* plyometric effect). The initial eccentric contraction experienced during landing forces a quick stretch of the muscles and tendons. As a defense mechanism against injury, the muscles involved will activate otherwise dormant muscle fibers in order to resist this stretching action, hence increasing recruitment. Additionally, if the athlete transitions immediately from the eccentric phase to the concentric, the muscle fibers involved are forced to contract very quickly, thus improving contraction speed.¹²

Plyometrics are used in climbing to improve recruitment, and more importantly, contact strength, the key to performing difficult, dynamic climbing moves, and thus hard routes or boulder prob-

lems. The challenge of applying plyometric training to climbing is that usually, finger contractions are isometric, whereas plyometric training requires the cycle of concentric to isometric to eccentric contractions. This challenge was overcome with the invention of the *Campus Board*.

“On Wallstreet there is a very difficult boulder problem.... Through the winter I went to this training center in Nuremberg, and with the help of a physical education professor I worked specific muscles, training intramuscular coordination combined with reaction time so that I would be able to do the moves for this problem.”

— Wolfgang Gullich⁶

THE CAMPUS BOARD

The incomparable Wolfgang Gullich was perhaps the first climber to recognize the power of plyometrics and apply it to climbing training. Gullich’s visionary adaptation of these concepts proved to be the key to his groundbreaking ascents of Wallstreet in 1987 (the world’s first 5.14b) and the legendary Action Directe in 1991 (5.14d, and still one of the hardest routes in the world).

To apply plyometric methods to his training, Gullich invented the now-infamous Campus Board. Gullich installed the original board at a Nuremberg fitness club known as “The Campus Centre” (hence the name), and used this tool to help elevate his power to levels that could only be described as “futuristic.”⁴ The board consisted of a ladder of one-pad-deep wooden finger edges fixed to a slightly overhanging wall. Gullich’s original method was to use momentum to move dynamically between these edges with feet dangling (jumping from handhold to handhold, so to speak, now known as *campusing*). Climbers have since developed countless variations on the basic exercise. Although the Campus Board is still the best medium for isolating plyometric movements, any difficult, dynamic movement (with or without feet on the wall) has the potential to create a plyometric effect.

POWER TRAINING METHODS

The ideal power-training exercise would increase motor-unit recruitment, increase fiber-contraction speed, and be sufficiently sport-specific to develop technical skill and specific body strength. Climbers can accomplish these objectives through a combination of the two fundamental power training methods described below: *limit bouldering* and *campusing*.

LIMIT BOULDERING

Limit bouldering is arguably the ideal method for improving power and contact strength while focusing attention on sport-specific factors like technique and body strength. Limit bouldering is climbing short boulder problems that feature one to two crux moves that are right at the climber's limit. This training activity is frequently confused with "bouldering," which entails climbing problems of any length with an emphasis on fun, camaraderie, and sending the hardest line possible. Limit bouldering, on the other hand, entails focusing on short boulder problems that emphasize one or two extremely hard moves (rather than problems that entail six to eight *pretty* hard moves, or 10 to 20 *kinda* hard moves). To improve recruitment and pow-



Limit boulder problems should feature holds that are specific to your climbing goals, and one or two dynamic moves that are right at your limit.

er, the number of repetitions must be small, and the intensity very high. Bouldering at a lower intensity is enjoyable and can be applied to other training goals, but for power training, the moves must be extremely difficult and few in number.

Limit bouldering is superior to other power-training methods because it involves the entire body in a highly sport-specific manner, it's easy to perform indoors or out, it includes skill practice, and it's fun. The downsides are that it's difficult to control and quantify (making it difficult to ensure progressive overload is achieved) and does not facilitate isolation of individual muscle groups (the forearm flexors in particular). Furthermore, it's easy to get sidetracked during limit-bouldering sessions, and it can be challenging to find suitable training terrain.

To get the most from this training, limit boulder problems should address specific weaknesses or areas of emphasis within the training plan. Utilize wall angles and hold types that are specific to your goals, be they crimps, pockets, pinches, etc. Keep in mind that the existing boulder problems in most commercial facilities emphasize big heroic moves between jugs, slopers, pinches, and other features that are not frequently encountered on real rock climbs. Likewise, the footholds are often much bigger than those found on real rock.

Emphasize hard, dynamic moves to achieve a plyometric effect and improve contact strength. Moves that require a precise deadpoint to a small target hold are ideal. If a move can be completed in three or four tries, it is not a limit move. A good limit boulder problem should take two or three *days* of effort to complete (with each session consisting of only five to ten serious attempts before fatigue prevents further progress). The moves involved should not be overly technical; the goal is to isolate recruitment and contraction speed — overly complicated moves only interfere with these objectives.

LIMIT BOULDERING ROUTINE

Like any training activity, begin with a thorough warm-up. The first phase of the warm-up should entail 10 to 30 minutes of low-intensity ARC-style traversing. Treat this period like any ARC set, focusing on efficient technique and smooth, relaxed movement. Near the end of this period do some active stretching while still on the wall.

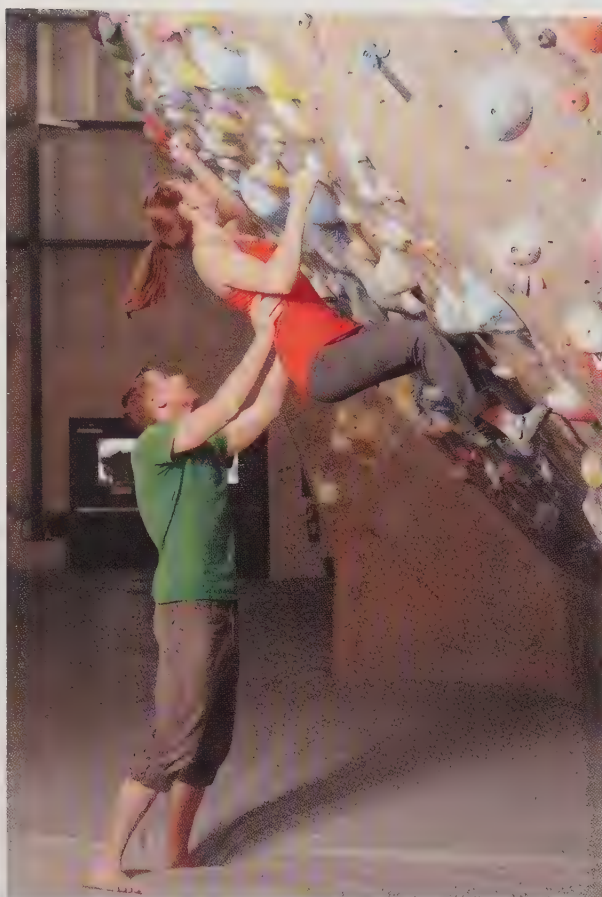
During phase two of the warm-up, complete a progressive Warm-Up Boulder Ladder (WBL). Begin with "easy" bouldering (starting at V0, or the easiest problems available). Complete one to four boulder

problems at each V-grade before progressing to the next grade (the number of problems completed at each grade depends on how many grades you need to step through, with the goal of completing the WBL in 20 to 30 minutes). Continue up the ladder until the typical boulder flash level is reached. The goal is *not* to get entrenched in an epic project during the warm-up phase. Climb a variety of problems that span multiple climbing styles. Take typical rest periods between problems (one to five minutes, increasing as the level of difficulty increases). By the end of the warm-up, you should have completed 10 to 20 problems of increasing difficulty.

At this point, the climber should feel warm, limber, and ready to crank, but she may not be warm enough to produce maximal performance. The neurological pathways that initiate motor-unit contractions require a thorough warm-up as well, so performance may continue to improve through the limit-bouldering session as these pathways perform more effectively.

Next complete 20 to 30 minutes of “hard” bouldering. Pick three to four problems harder than your flash level and work them for five to 10 minutes each. Ideally, these would be problems that require three to four attempts to climb. These problems should address certain areas of training emphasis (a weakness, or a climbing move replicating a goal route). On a good day, it should be possible to send most of these problems, but that is not the goal. It’s easy to get sidetracked during this activity, so watch the clock and stay focused on the big picture.

Next, move on to the limit boulder problems. These should be carefully selected (or better, *designed*) with specific attributes. In theory, these problems will be worked throughout most or all of the Power Phase, so they should be enjoyable and specific to the season goals. Two or three problems of the correct difficulty are usually enough to last through the phase. These problems should be right at the climber’s limit, and they should be powerful, with one or two extremely hard, dynamic moves that she can’t do (as opposed to 10 consecutive *pretty* hard moves that result in a pump-management challenge). If necessary, consider using a power spot to work these moves initially. The crux should be low to the ground (but probably not the first move) so that the climber is fresh for each attempt at the crux, and fear is not a factor. Ideally, the crux moves will have a dynamic element, requiring precision at the end of a long reach and good contact strength to latch the finishing hold. Try to keep these problems as realistic as possible, using small footholds



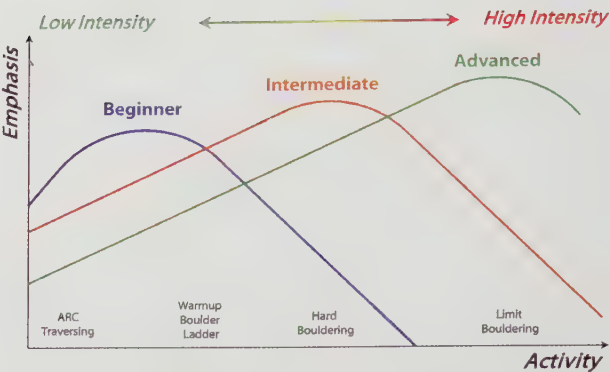
A “Power Spot” can be used to train limit moves. Ask your partner to exert a small amount of upward force just above your hips, and to gradually reduce the assistance provided as you improve at the move.

that require precise footwork and core tension, and handholds that mimic specific goal routes. The set of problems selected should be diverse, covering the different styles needed to achieve all of the climber’s goals.

Spend 15 minutes (or more) on each limit boulder problem. Ideally, you’ll make at least four or five solid attempts, with sufficient rest to ensure maximum effort is applied to each attempt. It’s not necessary to work each problem or dedicate equal time to each problem during a given session. Rest for five to 10 minutes between problems.

At some point during the limit-bouldering period power will start to wane. End the workout as soon as it’s no longer possible to move explosively. There is nothing more to be gained (in terms of power) at this point. Continuing with easier problems only increases the risk of injury and the time required to recover from the workout. Finish the workout with two to four supplemental exercises (perform the set and repetition protocols listed for “Power” in the Table on p110 in [Chapter 6: Strength](#)).

The limit-bouldering routine should be tailored to the climber’s ability level and training experience by adjusting the duration and intensity of the various phases of the workout. Less-experienced climbers will achieve significant physical improvements with relatively little training, and can benefit most from skill-specific training.³ Therefore, these climbers should spend most of their time on the first few phases of the workout (ARC traversing, WBL, and, to a lesser extent, hard bouldering), striving to complete a relatively high volume of challenging movements. Since a beginner’s ladder won’t be quite as “tall” as an advanced climber’s, beginners can perform more problems on each rung of the ladder, thus experiencing a wider variety of moves. More-advanced climbers should spend more time attempting the most difficult moves, thus applying more time to the latter part of the workout, as described in the Figure to the right.



Less-experienced climbers should focus more effort on the less intense parts of the Limit Bouldering Routine, and vice versa for more-advanced climbers.

LIMIT BOULDERING ROUTINE

WARM-UP:

	Duration of Activity		
	Beginner	Intermediate	Advanced
Phase 1: Low-intensity ARC traversing	30 min.	20 min.	10 min.
- Rehearse techniques - Light, active stretching on the wall			
Phase 2: Warm-Up Boulder Ladder	30 min.	25 min.	25 min.
- Begin with “V-easy” 1 to 4 problems of each grade up to flash level - No more than three attempts per problem - Increase rest periods with difficulty	4 problems /grade	3 problems /grade	2 problems /grade
Phase 3: Hard Bouldering	30 min.	30 min.	25 min.
3 to 4 problems just above flash level - Problems should address areas of training emphasis 3 to 4 attempts per problem	3 problems	4 problems	3 problems

LIMIT BOULDERING:

	Duration of Activity		
	Beginner	Intermediate	Advanced
2 to 4 “realistic” problems - Feature 1 or 2 powerful moves that you cannot do in a single session of work - Dynamic cruxes, low to the ground - Four or five quality attempts per problem - Extremely high intensity and arousal 2 to 5 minutes rest between attempts, 5 to 10 minutes rest between problems	0 min. 0 problems	30 min. 2 problems	60 min. 4 problems

SUPPLEMENTAL EXERCISES:

- Select one to three high-priority movements
- Complete one to three sets of each exercise

The Limit Bouldering Routine includes activities of various intensity. Adjust the emphasis of Limit Bouldering sessions to your abilities.

"I like to do Campus Board, because it is very effective and you don't need a lot of time for that." — Iker Pou¹

CAMPUS TRAINING

This dynamic style of training involves footless dynos between like holds, and is probably the best method of pure power training available to climbers due to its plyometric nature. Unlike limit bouldering, campusing is relatively easy to quantify, so climbers can track progress from workout to workout and year to year. Campusing also permits better isolation of specific grip positions than limit bouldering.

In addition to the pure power benefits of campus training, this method is very helpful for improving the intramuscular coordination required to make accurate dynamic moves. Practicing dynos, or campus, moves improves spatial awareness and motor skills, resulting in improved "aim" when launching toward a target hold. In just a few sessions it's possible to prog-

ress to a point where virtually every campus move is a "deadpoint." This accuracy translates directly to the rock, a skill element of campus training that is often overlooked. The campus board is an extremely powerful tool for teaching climbers of *all abilities and experience levels* to move with speed and accuracy.

Finally, it's well known that some climbers struggle with dynamic moves. This could be due to a general lack of aggression or a strong desire to remain "in control" on the rock. Campus training can solve these issues in short order. It demands aggressive and committing movement, in a structured, low-risk environment. Numerous climbers have corrected years of overly static movement after only a handful of short campus sessions.

With all the many benefits of campusing, it's worth noting the downsides. First, there is no doubt that campusing is much harder on the joints than other training methods, such as hangboarding. Campusing is, by its very nature, somewhat wild and out of control. A hangboard allows the climber to adjust the intensity at will, and let go the instant a repetition becomes uncomfortable. Often while campusing (or dynoing in general), the only sign of injury comes after it's too late. For that reason, it's critical to minimize the amount of time spent on the campus board, and ensure that the climber is 100 percent injury-free before beginning any campus activities. Elbows are particularly at risk, but shoulders and fingers must be healthy as well. Campus training should only be performed with an open-hand or half-crimp-grip position (index and pinkie fingers straight, middle and ring fingers bent approximately 90 degrees at the big knuckle). Do not campus with a closed crimp grip. Finally, campusing can be hard on the finger pads. Pay attention to skin health and keep workouts short and intense to limit skin degradation.

Campus training requires a thoughtfully designed Campus Board, which many gyms don't have. Fortunately building a board is relatively simple and inexpensive (see the sidebar on p136). Campusing can be done on a regular bouldering wall, systems wall, or similar wall with a series of regularly spaced, like holds, but a proper Campus Board is preferred.

A good board will have several different sizes of wooden "rungs" (plastic rungs are hard on the finger pads). Err on the side of using the smallest manageable rungs. Smaller rungs will isolate finger strength, while larger (deeper) rungs isolate the upper arms, shoulder, and back muscles. Most climbers should focus on increasing finger strength rather than big-mus-



Consistent Campus Training will greatly improve inter-muscular coordination, key for moves requiring dynamic accuracy like this dyno to a pocket.

cle strength. The large rungs are helpful for warming up and learning the basic movements, but climbers should progress to the smallest rungs as soon as possible.

One final note: Campusing can be extremely fun, bordering on addictive, to the point that it has the po-

tential to become a major distraction. It's not uncommon for serious campus enthusiasts to lose focus on the big picture in order to achieve a specific campus goal (like what Ben Moon calls the "ultimate expression of power," the 1-5-9 ladder). Enjoy campusing but don't get sidetracked.

The Campus Board

Campus Boards come in many shapes and sizes, and not all are created equal. A well-constructed board will make workouts more enjoyable and facilitate quick and steady progression. The board must be securely attached to a solid structure, as it will be subjected to heavy dynamic forces. There are several primary variables in Campus Board design:

- Rung type and material
- Rung size (depth)
- Rung spacing
- Board angle

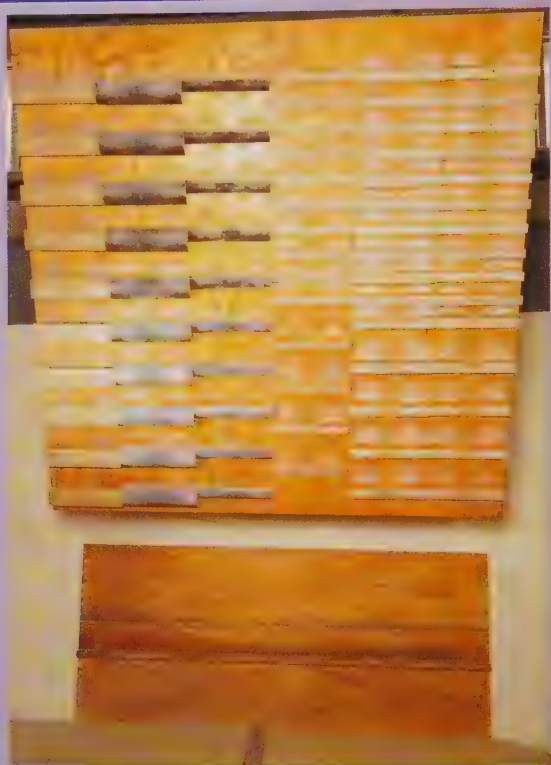
Wood is the material of choice for campusing connoisseurs. It offers the right amount of friction and durability. Rungs can be flat or slightly incut, but they must be skin friendly, with a well-rounded and smooth leading edge. Each rung should be numbered so that you can record and track your training. Metolius makes three sizes of wooden campus rungs that are highly recommended ($\frac{3}{4}$ ", 1", and 1- $\frac{1}{4}$ "-deep). They also have a free download on their website describing how to construct a board.

Rung size (or depth) can, and should, vary. Beginners can use deeper rungs, while more-advanced climbers should use shallower rungs. A good board will have a variety of rungs (usually two to three sizes) but these rungs should NOT be mixed in the same ladder.

Standard rung width is 16", allowing three distinct columns of rungs to fit side-by-side onto one four-foot-wide plywood sheet. Each rung in a single ladder should be the same size, shape, texture, and orientation. This uniformity provides the necessary repetition to isolate weaknesses and progress steadily.

Vertical spacing between rungs is a hotly debated topic. There are currently two competing standards. "Moon-spacing" (so-named for legendary climber and campus-fiend Ben Moon) is the most common outside the US, with rungs spaced 22 cm (8.66") apart *on-center* (meaning, the vertical distance from the center of one rung to the center of the next rung is 22 cm). "Metolius-spacing" is 4" on-center within a column of shallow-depth rungs, 6" on-center within a column of medium rungs, and 12" on-center within a column of large rungs. Tighter vertical spacing is preferred because it allows smaller increments of progression. If using Moon-spacing, alleviate this problem by adding "half-steps" (i.e., spacing rungs vertically at 11-cm on-center; this is still a bit wider, and therefore harder, than Metolius-spacing). Using standardized spacing allows you to compare your campus feats to others' around the world. Moon-spacing is gaining ground in the US, and will likely be the world standard moving forward.

Finally, not much thought is given to the steepness of the board, but it's an important factor. Steeper (more overhanging) boards require less vertical movement to make larger moves, meaning the effect of gravity is slightly reduced. Of course, it can be more difficult to grasp the rungs on a steeper board, since the rungs will seem more slopey. Those with experience on many boards report steeper boards are easier to use, up to a point. While there is no clear standard, 15 to 20 degrees overhanging is most common. Anything below 10 degrees increases the risk of inadvertently kicking the board, and puts extra stress on the elbows.





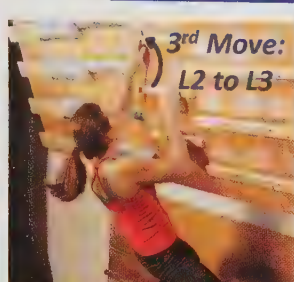
Start:
B1



1st Move:
L1 to L2



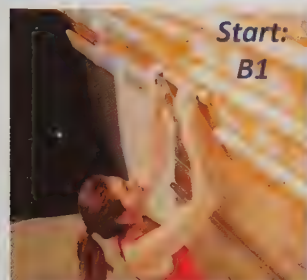
2nd Move:
R1 to R2



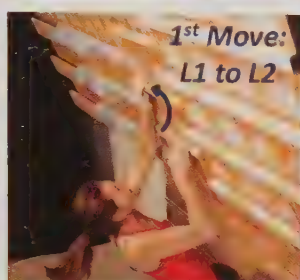
3rd Move:
L2 to L3

Matching Ladders: Climb the board one rung at a time, matching on every rung until reaching the top. For example, begin with both hands matched on rung No. 1, then bump the left hand to rung No. 2, then bump the right hand from rung No. 1 to rung No. 2, then the left hand to rung No. 3, and so on. Alternate sets leading with each hand.

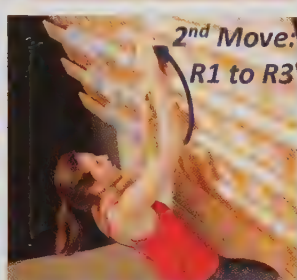
Note: B = Both Hands; L = Left Hand; R = Right Hand; Numerals indicate Campus Rung Number



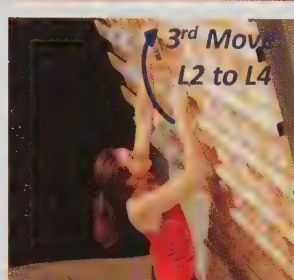
Start:
B1



1st Move:
L1 to L2



2nd Move:
R1 to R3



3rd Move:
L2 to L4

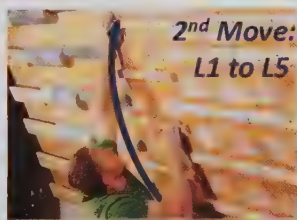
Basic Ladders: Climb the board one rung at a time, alternating hands, latching every rung until reaching the top. For example, begin with both hands on rung No. 1, then bump the left hand from rung No. 1 to rung No. 2, then move the right hand from rung No. 1 to rung No. 3 (without matching on rung No. 2), then move the left hand from rung No. 2 to rung No. 4, and so on, until the top of the board is reached.



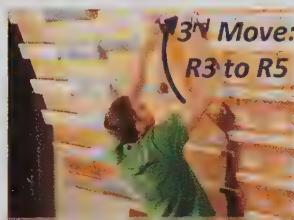
Start:
B1



1st Move:
R1 to R3



2nd Move:
L1 to L5

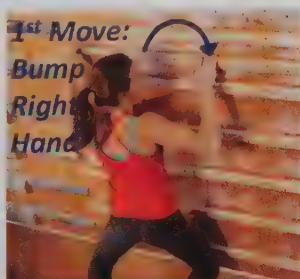


3rd Move:
R3 to R5

Max Ladders: Climb the board alternating hands, skipping as many rungs as possible. For example, start with both hands on rung No. 1, throw the right hand from rung No. 1 to rung No. 3, then, without matching, throw the left hand from rung No. 1 to rung No. 5, then match the right hand to rung No. 5. The ultimate goal is to travel as far up the board as possible in two hand moves. In this example, the climber goes from a match on rung No. 1 to a match on rung No. 5 (traveling four increments). This is the single best campus exercise for climbers. This is the most basic movement, the most specific to rock climbing, and the best for isolating individual hands.



Start:
B5



1st Move:
Bump
Right
Hand



2nd Move:
Bump RH
again



3rd Move:
Match

Typewriters: In this exercise, the climber campuses horizontally. Begin with hands matched on the left end of the board. Bump the right hand progressively farther right, without going up the board, until near failure or you run out of real estate. Then progressively bump the left hand to the right, eventually matching the left hand next to the right. Then reverse the exercise. These should be used infrequently to train for a specific move, and can be hard on the shoulders.

CAMPUS EXERCISES

A variety of campus exercises are described below, listed in approximate order of difficulty from easiest to hardest. These should be performed with feet dangling (though it's OK to use your lower body to help generate momentum, also known as *kiping*). If these exercises are too difficult, consider using a power spot to practice the movements. Use the smallest rungs manageable to emphasize the forearm flexors.

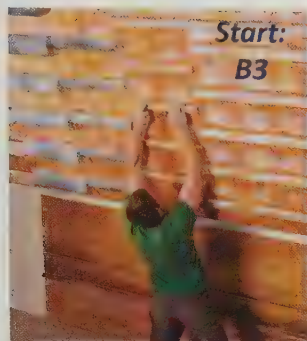
Any of these exercises can be done in both directions — that is, going up the board or going down. Downward campusing is known as *reactive* training because it activates the muscles' inherent injury-defense mechanism to recruit otherwise dormant fibers. Theoretically, it will produce more-dramatic results, but also increases injury risk significantly, because the forces on the elbow, shoulder, and finger joints are much higher. Keep in mind that down-campusing completely eliminates what little specificity exists in campus training. For these reasons, down-campusing should be reserved for highly trained athletes looking to push just a little bit harder (as a rule of thumb, those with five or more years of campus training under their belt). The weak link in down-campusing is almost always finger strength, so it can be a good way to isolate this area if other factors prevent doing so with more basic upward movements.

- **Matching Ladders:** Climb the board one rung at a time, matching on every rung until reaching the top. For example, begin with both hands matched on rung No. 1, then bump the left hand from rung No. 1 to rung No. 2, then bump the right hand from rung No. 1 to rung No. 2, then the left hand to rung No. 3, and so on. Alternate sets leading with each hand.

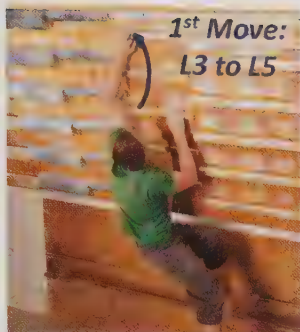


Two acceptable campus grip positions: open-hand with inactive pinky (L), and semi-closed crimp (R).

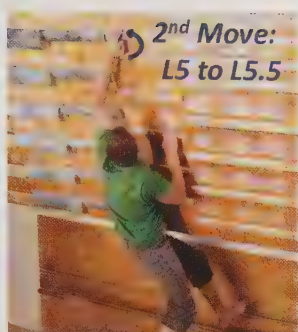
- **Basic Ladders:** Climb the board one rung at a time, alternating hands, latching every rung until reaching the top. For example, begin with both hands on rung No. 1, then bump the left hand from rung No. 1 to rung No. 2, then move the right hand from rung No. 1 to rung No. 3 (without matching on rung No. 2), then move the left hand from rung No. 2 to rung No. 4, and so on, until the top of the board is reached.
- **Max Ladders:** Climb the board alternating hands, skipping as many rungs as possible. For example, start with both hands on rung No. 1, throw the left hand from rung No. 1 to rung No. 5, then, without matching, throw the right hand from rung No. 1 to rung No. 9, then match the left hand to rung No. 9. The ultimate goal is to travel as far up the board as possible in two hand moves. In the above example, the climber goes from a match on rung No. 1 to a match on rung No. 9 (traveling eight increments). This is the single best campus exercise for climbers. This is the most basic movement, the most specific to rock climbing, and the best for isolating individual hands.
- **Typewriters:** In this exercise, the climber campuses horizontally. Begin with hands matched on the left end of the campus board. Bump the right hand progressively farther right, without going up the board, until near failure or until you run out of real estate. Then progressively bump the left hand to the right, eventually matching the left hand next to the right. Then reverse the exercise. These should be used infrequently to train for a specific move, and can be hard on the shoulders.
- **Max First Move:** Start matched on the first rung and throw one hand to the highest rung possible (leaving the other hand on the first rung). These are very specific to rock climbing, but also good for building the shoulder strength necessary for progressing on max ladders. Once the high rung is latched, try to match (or pass the high rung).
- **Go-Agains (aka "Bumps"):** Complete the first move of the max-ladder exercise, then continue bumping the high hand one rung at a time until failure, leaving the low hand on the first rung. This exercise can be hard on the shoulders.



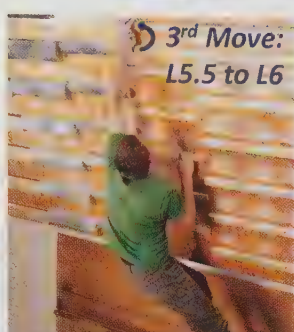
Start:
B3



1st Move:
L3 to L5

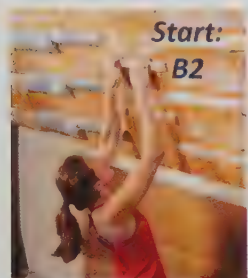


2nd Move:
L5 to L5.5



3rd Move:
L5.5 to L6

Go-Agains (aka "Bumps"): Complete the first move of the max-ladder exercise, then continue bumping the high hand one rung at a time until failure, leaving the low hand on the first rung. This exercise can be hard on the shoulders.



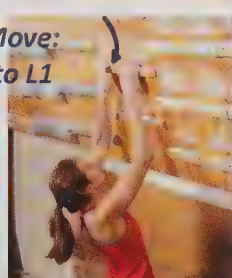
Start:
B2



1st Move:
L1 to touch L2



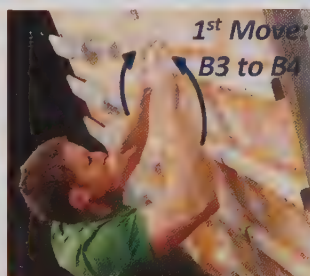
2nd Move:
L2 to L1



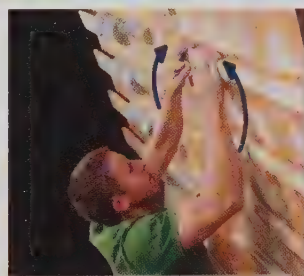
Touches: Begin as for the first move of a max ladder, but instead of latching the high rung, simply touch the rung, then fall slowly back onto the starting rung. These work upper-arm explosive power and lock-off strength in a "negative" eccentric contraction.



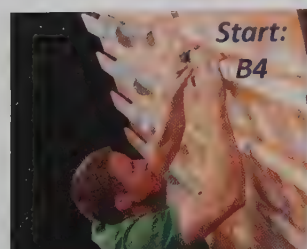
Start:
B3



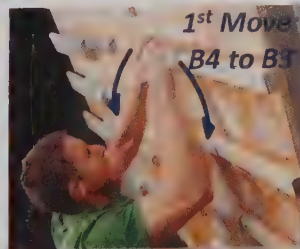
1st Move:
B3 to B4



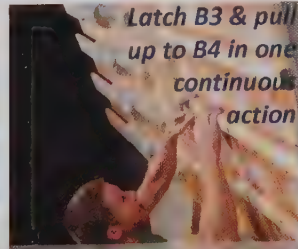
Double Dynos (aka "Doubles"): Move both hands at the same time in matched double-dynamic fashion — so for each movement you'll have no contact with the board. These lack specificity, but they are also the most plyometric and are great for developing dynamic aggression, spatial awareness, and hand-eye coordination at high speeds. Additionally, doubles require a quick latch of the target rung, thus improving contact strength.



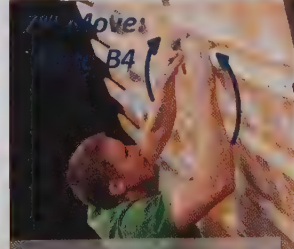
Start:
B4



1st Move:
B4 to B3



Latch B3 & pull up to B4 in one continuous action



2nd Move:
B4

Up-Down-Up: A more advanced, plyometric version of the other exercises, usually done with a double dyno, but can be done with one hand fixed if doubles prove too hard. Begin on a relatively high rung, then down-campus a set increment. When the lower rung is latched, explode back up to the starting rung, striving to make the down-and-up portion one continuous movement, executed as quickly as possible. These emphasize the plyometric cycle of eccentric/concentric contraction in the upper arms, though they also significantly increase the load on the fingers.

- **Touches:** Begin as for the first move of a max ladder, but instead of latching the high rung, simply touch the rung, then fall slowly back onto the starting rung. These work upper-arm explosive power and lock-off strength in a “negative,” eccentric contraction.
- **Double Dynos (aka “Doubles”):** Move both hands at the same time in matched double-dynamic fashion — so for each movement you’ll have no contact with the board. These are among the least specific campus exercises (campus expert Ben Moon dislikes doubles, citing this lack of specificity),⁸ but they are also the most plyometric and are great for developing dynamic aggression, spatial awareness, and hand-eye coordination at high speeds. Additionally, doubles

require a quick latch of the target rung, thus improving contact strength.

- **Up-Down-Up:** This is a more advanced plyometric version of the other exercises, and is usually done with a double dyno, but can also be done with one hand fixed if doubles prove too hard (as for max ladders). Begin on a relatively high rung, then down-campus a set increment (with a single hand or matched hands). When the lower rung is latched, explode back up to the starting rung, striving to make the down-and-up portion one continuous movement, executed as quickly as possible. These emphasize the plyometric cycle of eccentric/concentric contraction in the upper arms, though they also significantly increase the load on the fingers.

Tips for Effective Campusing

There’s more to campusing than bulging back muscles and misplaced aggression. Campusing is a skill that improves with practice. Campusing with proper technique will allow you to perform better on the board, surely, but it will also help reduce injury risk and achieve the most sport-specific power gains possible. These tips will help you get the most benefit from your campus sessions:

- Dynamic movements depend on effective use of momentum. The same is true of campusing. Try to keep your center of mass moving continuously up the board, with each move flowing rhythmically into the next. Avoid pausing or adjusting between moves. This will keep the exercises plyometric.
- Keep your eyes wide open and focused on your target throughout the movement, but especially while latching the rung. Many climbers inexplicably close their eyes during dynamic movements. Use the campus board to break this undesirable habit.
- Practice deadpointing each move. At first, this will take a great deal of control and concentration, but with practice you will be able to do this on moves right at your limit. This skill transfers directly to the rock and will save your elbows and fingers tremendous strain.
- On big moves, focus on pushing with your lagging hand. Climbers are used to pulling, and once a hand drops below shoulder level, it is often forgotten. Get in the habit of pushing down with your low hand all the way through the movement.
- Twist in toward the lagging hand to get a bit of extra reach, just as when performing a drop-knee or twist-lock with the feet.



To maximize reach and efficiency during extended campus moves, rotate the lagging shoulder into the board and use your triceps to push down on the low rung with the lagging hand.

- Three-/Two-/One-Finger Campusing:** For the hardcore, any of the exercises described above can be performed with three, two, or even one finger. Campusing in this manner isolates finger strength.

A common mistake is trying to do too many different exercises on the campus board in one session. To achieve the desired power-training objective, a campus workout must be short and intense, so there isn't time to properly execute all of these exercises. Remember, these movements are supposed to be right at your physical limit. If you can do an exercise on the first attempt, it is not limiting. Instead, focus on a handful of limiting exercises and attempt them repeatedly to allow progression to occur. It should take many, many tries over multiple sessions to succeed on the most powerful movements.

BEGINNER CAMPUS ROUTINE

Note: This training regimen is for the thoroughly healthy. Climbers with nagging injuries, particularly finger, elbow, or shoulder injuries, SHOULD NOT DO THIS! Use another training method, or consider less-intense goals until healthy. The frequency and rest associated with these workouts are critical to avoiding injury. Avoid consecutive campus workouts, and take extra rest following each campus workout.

This routine is an introduction to campus training designed for those who are new to structured campusing. This routine is focused on learning the basic movements, practicing dynamic accuracy, and overcoming inhibitions. Emphasize smooth

movement, using momentum to flow from one move to the next. Focusing on mastering relatively easier ladders, as opposed to flailing on more difficult ones, will provide the quickest improvement. Once all of the exercises in this routine can be performed, transition to the Intermediate Campus Routine.

Campusing should only be done after a thorough warm-up. Additionally, campusing should also be done early enough in the workout that the climber still feels "explosive" and able to "pop" off of every rung. Warm-up as described for the Limit Bouldering Routine, completing a 20-minute ARC set, then a 30-minute Warm-Up Boulder Ladder, followed by 10 minutes of Hard Bouldering. Record the results of each set in a log sheet like the one shown below.

Use a log sheet to document your campus training sessions.

Campus Log Sheet

Campus Board Configuration:

Date: 15 Feb 2011

W/O #: 4

Bouldering Summary/Notes:

Bouldered 55 minutes, 2 x V1 thru V6.

Sent 2xV8, Flashed 1xV7. Worked V9 for 10 minutes.

R = Right Hand, L = Left Hand, B = Both Hands

Rungs Used (circle):

Basic Ladders	Set 1:	(L) M S	B1-L2-R3-L4-R5-B5; B1-R2-L3-R4-L5-B5
	Set 2:	(M) S	B1-L3-R5-L7-R9-B9; B1-R3-L5-R7-L9-B9
	Set 3:	L M S	B1-L4-R7-L10-R13-B13; B1-R4-L7-R10-L13-B13
Max Ladders	Set 4:	L M S	B1-L7-R12-B12; B1-R7-L12-B12
	Set 5:	L M S	B1-L7-R13/Off; B1-R7-L13/Off
	Set 6:	L M S	B1-L7-R13-B13; PR! B1-R7-L13-B13 PR!
	Set 7:	L M S	B1-L7-R13-B13; B1-R7-L13-B13
	Set 8:	L M S	B1-L8-R13/Off; B1-R8-L13/Off
	Set 9:	L M S	B1-L8-R13-B13; B1-R8-L13/Off
Warmup Doubles	Set 10:	(L) M S	B1-B2-B3-B4-B5
	Set 11:	(M) S	B1-B3-B5-B7-B9
	Set 12:	L M S	B1-B4-B7-B10-B13
Doubles Dynos	Set 13:	L M S	B1-B5-B9
	Set 14:	L M S	B1-B6-B11
	Set 15:	L M S	B1-B7-B12/Off
	Set 16:	L M S	B1-B7-B12
	Set 17:	L M S	
	Set 18:	L M S	

Great session! 30 min Total, new Personal Record on Max Ladder!

BEGINNER CAMPUS ROUTINE

WARM-UP:

Phase 1: Low-intensity ARC traversing (20 Minutes)

- Rehearse techniques
- Light, active stretching on the wall

Phase 2: Warm-Up Boulder Ladder (30 Minutes)

- Begin with "V-easy," complete 1 to 3 problems of each grade up to flash level
- No more than three attempts per problem

Phase 3: Hard Bouldering (10 minutes)

- 2 "realistic" problems
- 3 to 4 quality attempts per problem
- 2 to 5 minutes rest between attempts

CAMPUS EXERCISES:

Notes:

- Assumes "Metolius" rung spacing or "Moon" half-spacing:
- Key: **B** = both hands, **L** = left hand, **R** = right hand, number indicates rung number

Matching Ladders

- | | |
|-----------------------|----------------------|
| • Set 1: Large Rungs | B1-L2-R2-L3-R3-L4-B4 |
| • Set 2: Large Rungs | B1-R2-L2-R3-L3-R4-B4 |
| • Set 3: Medium Rungs | B1-L2-R2-L3-R3-L4-B4 |
| • Set 4: Medium Rungs | B1-R2-L2-R3-L3-R4-B4 |

Basic Ladders

- | | |
|-----------------------|----------------------|
| • Set 5: Large Rungs | B1-L2-R3-L4-R5-L6-B6 |
| • Set 6: Large Rungs | B1-R2-L3-R4-L5-R6-B6 |
| • Set 7: Medium Rungs | B1-L2-R3-L4-R5-L6-B6 |
| • Set 8: Medium Rungs | B1-R2-L3-R4-L5-R6-B6 |

Max Ladders - Summon maximal effort and arousal for each attempt

- | | |
|--|---------------------|
| • Set 9: Medium Rungs | B1-L3-R4-B4 |
| • Set 10: Medium Rungs | B1-R3-L4-B4 |
| • ...Repeat as necessary until successful | |
| • Set 11: Medium Rungs | attempt B1-L3-R5-B5 |
| • Set 12: Medium Rungs | attempt B1-R3-L5-B5 |
| • ...Repeat until successful, but no more than eight total sets of Max Ladders | |

SUPPLEMENTAL EXERCISES:

- 2 to 4 exercises addressing specific physical weaknesses
- 1 to 3 sets each of 8 or fewer repetitions

NOTES ON BEGINNER CAMPUS ROUTINE

- Most beginners will find it difficult to use the Campus Board at first. If possible, get a power spot from a partner. Reduce the support from the power spot as you progress.
- Begin with the largest or easiest rungs, and then progress to smaller rungs. Depending on the board's rung spacing, medium rungs may be easier than large rungs, etc.
- Do not attempt max ladders until basic ladders can be performed smoothly, with good control.
- Emphasize smooth, flowing movement over attempting big moves.

INTERMEDIATE CAMPUS ROUTINE

Note: This training is for the thoroughly healthy. Climbers with nagging injuries, particularly finger, elbow, or shoulder injuries, SHOULD NOT DO THIS! Use another training method, or consider less-intense goals until healthy. The frequency and rest associated with these workouts are critical to avoiding injury. Avoid consecutive campus workouts, and take extra rest following each campus workout.

This routine is intended for climbers who can smoothly and reliably complete basic ladders without assistance. Unlike the Beginner Campus Routine, this workout includes greater emphasis on physical power, as opposed to movement skill and commitment. However, you should continue to focus on completing the moves smoothly, using momentum to flow from one move to the next. Focusing on mastering relatively easier ladders, as opposed to flailing on more-difficult ones, will provide the quickest improvement. Once all of the exercises in this routine can be performed smoothly, transition to the Advanced Campus Routine.

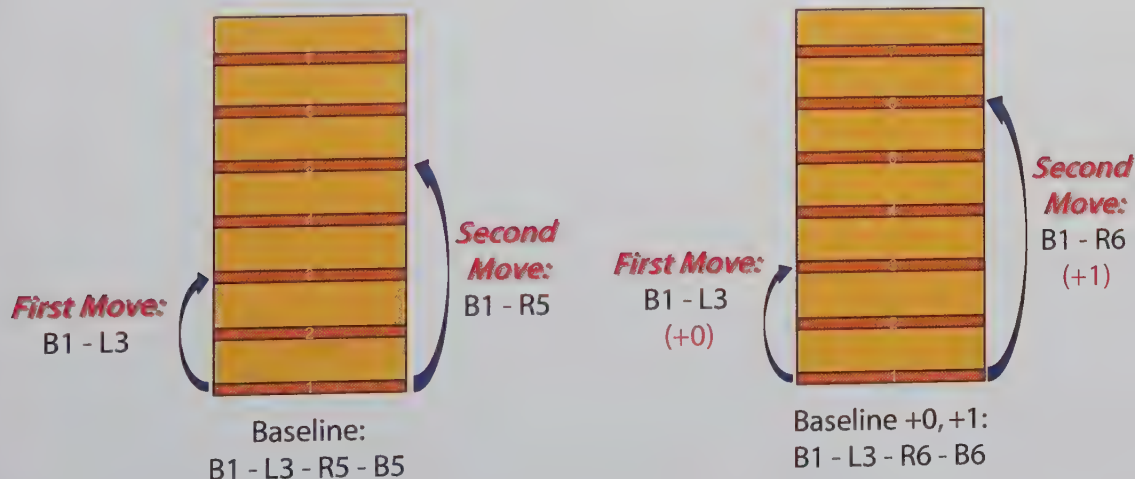
Campus should only be done after a thorough warm-up. Additionally, campus should also be done early enough in the workout that you still feel “explosive” and able to “pop” off of every rung. Warm-up as described for the Limit Bouldering Routine, completing a 15-minute ARC set, then a 30-minute Warm-Up Boulder Ladder, followed by 15 minutes of Hard Bouldering. Record the results of each set in a log sheet like the one shown in the Figure on p141.

NOTES ON INTERMEDIATE CAMPUS ROUTINE

- Campus exercises are denoted as “Baseline +X, +Y”, which describes each set relative to the climber’s “baseline” max ladder. The base-

line max ladder is the most difficult ladder the climber can reliably execute, for example, small rungs, B1-L3-R5-B5. The next increment of difficulty would be B1-L3-R6-B6, and is denoted as “Baseline +0, +1”, as shown in the figure below. This numbering refers to the rung increments spanned during each move of the ladder, relative to the climber’s baseline. The first move deposits the climber’s left hand on rung 3, the same rung reached during the first move of the baseline ladder, so “Baseline +0” is used to denote the first move. The second move deposits the climber on rung 6, one rung higher than the rung reached during the second move of the baseline ladder, so “+1” is used to denote the second move of this ladder. The final move of the max ladder is always a match to the rung reached during the second move, so it is omitted.

- Depending on the board’s rung spacing, it may be prudent to skip a rung or two during basic ladders.
- Once warmed up, perform all sets on the smallest rungs manageable.
- Perform no more than 12 total sets of max ladders (often much less than 12 is best). Begin with a baseline max ladder, such as small rungs, B1-L3-R5-B5. Use the remaining sets to increase the spacing of the ladder. For example, attempt small rungs, B1-L3-R6-B6, etc.
- Double dynos will be difficult at first, so consider using a power spot. Do not attempt double dynos until max ladders can be performed smoothly. At first, use the largest or easiest rungs available, and do not attempt to skip any rungs.
- When performing double dynos, emphasize the plyometric cycle by latching and moving off each rung as quickly as possible.



INTERMEDIATE CAMPUS ROUTINE

WARM-UP:

Phase 1: Low-intensity ARC traversing (15 Minutes)

- Rehearse techniques
- Light, active stretching on the wall

Phase 2: Warm-Up Boulder Ladder (30 Minutes)

- Begin with "V-easy," complete 1 to 2 problems of each grade up to flash level
- No more than three attempts per problem

Phase 3: Hard Bouldering (15 minutes)

- 2 "realistic" problems
- 3 to 4 quality attempts per problem
- 2 to 5 minutes rest between attempts

CAMPUS EXERCISES:

- Assumes "Metolius" rung spacing or "Moon" half-spacing:
- Key: **B** = both hands, **L** = left hand, **R** = right hand, number indicates rung number

Basic Ladders

- Set 1: Large Rungs B1-L2-R3-L4-R5-L6-B6
- Set 2: Large Rungs B1-R2-L3-R4-L5-R6-B6
- Set 3: Medium Rungs B1-L2-R3-L4-R5-L6-B6
- Set 4: Medium Rungs B1-R2-L3-R4-L5-R6-B6
- Set 5: Small Rungs B1-L2-R3-L4-R5-L6-B6
- Set 6: Small Rungs B1-R2-L3-R4-L5-R6-B6

Max Ladders - Summon maximal effort and arousal for each attempt; *THIS IS WAR!*

- Set 7: Small Rungs Baseline Max Ladder (leading with left hand)
- Set 8: Small Rungs Baseline Max Ladder (leading with right hand)
- Set 9: Small Rungs Baseline +0, +1 Max Ladder (leading with left hand)
- Set 10: Small Rungs Baseline +0, +1 Max Ladder (leading with right hand)
- ...Repeat until successful, but no more than 12 total sets of Max Ladders

If successful...

- Sets 11-14: Small Rungs Progressive Max Ladders (alternate leading hand)
- ...Repeat until successful, but no more than 12 total sets of Max Ladders

Double Dynos

- Set 15: Medium Rungs B1-B2-B3-B4
- Set 16: Small Rungs B1-B3-B5 (or Baseline Double Dyno)
- Set 17: Small Rungs Baseline +1, +1
- Set 18: Small Rungs Baseline +1, +2
- ...Repeat above sets until successful, but no more than four total sets of Doubles

SUPPLEMENTAL EXERCISES:

- 2 to 4 exercises addressing specific physical weaknesses
- 1 to 3 sets each of 8 or fewer repetitions

ADVANCED CAMPUS ROUTINE

WARM-UP:

Phase 1: Low-intensity ARC traversing (10 Minutes)

- Rehearse techniques
- Light, active stretching on the wall

Phase 2: Warm-Up Boulder Ladder (30 Minutes)

- Begin with "V-easy," complete 1 to 3 problems of each grade up to flash level
- No more than three attempts per problem

Phase 3: Hard Bouldering (20 minutes)

- 2 to 3 "realistic" problems
- 3 to 4 quality attempts per problem
- 2 to 5 minutes rest between attempts

CAMPUS EXERCISES:

- Assumes "Metolius" rung spacing or "Moon" half-spacing:
- Key: **B** = both hands, **L** = left hand, **R** = right hand, number indicates rung number

Warm-up Ladders

- Set 1: Medium Rungs B1-L3-R5-L7-R9-L10-B10
- Set 2: Medium Rungs B1-R3-L5-R7-L9-R10-B10
- Set 3: Small Rungs B1-L4-R7-L10-R13-L16-B16
- Set 4: Small Rungs B1-R4-L7-R10-L13-R16-B16

Max Ladders - Summon maximal effort and arousal for each attempt; *THIS IS WAR!*

- Sets 5-6: Small Rungs Baseline Max Ladder (leading with each hand)
- Sets 7-8: Small Rungs Baseline +0, +1 Max Ladder (leading with each hand)
- Sets 9-20: Small Rungs Progressive Max Ladders (alternate leading hand)
- ...Repeat until successful, but no more than 16 total sets of Max Ladders

Double Dynos

- Set 21: Medium Rungs B1-B3-B5-B7
- Set 22: Small Rungs Baseline Double Dyno
- Set 23: Small Rungs Baseline +1, +1
- Sets 24-5: Small Rungs Progressive Double Dynos
- ...Repeat above sets until successful, but no more than four total sets of Doubles

Up-Down-Up

- Set 25: Small Rungs B2-B1-B2
- Set 26: Small Rungs B3-B1-B3
- Sets 27-28: Small Rungs Progressive Up-Down-Up

SUPPLEMENTAL EXERCISES:

- 2 to 4 exercises addressing specific physical weaknesses
- 1 to 3 sets each of 8 or fewer repetitions

ADVANCED CAMPUS ROUTINE

Note: This training is for the thoroughly healthy. Climbers with nagging injuries, particularly finger, elbow, or shoulder injuries, SHOULD NOT DO THIS! Use another training method, or consider less-intense goals until healthy. The frequency and rest associated with these workouts are critical to avoiding injury. Avoid consecutive campus workouts, and take extra rest following each campus workout.

This routine is for well-trained climbers with extensive experience using the campus board. This routine introduces extremely intense reactive movements, so climbers should have approximately five years of campus training experience before attempting it. This experience is necessary to know how hard to push and when to stop.

Warm-up as described for the Limit Bouldering Routine, completing a 10-minute ARC set, then a 30-minute Warm-Up Boulder Ladder, followed by 20 minutes of Hard Bouldering. Record the results of each set in a log sheet like the one shown in the Figure on p141.

NOTES ON ADVANCED CAMPUS ROUTINE

- Up-down-ups will be very difficult initially. Use a power spot or foot dab, and perform on larger rungs before progressing to smaller rungs and larger moves.
- To get the most value out of up-down-ups, the period of transition from latch to upward motion must be as brief as possible. Favor shorter drops with a quicker transition over longer drops with a slower transition.

FOCUS AND INTENSITY

Effective power training demands high intensity. Whether bouldering or campusing, each problem or set (excluding warm-up activities) should include one or two limit moves. Such moves can only be completed with maximal effort. Summoning maximal effort many times within a single session is not trivial. It requires focus, attention, and strong motivation. Fortunately, this process becomes easier with practice, and translates directly to performance opportunities on the rock.

Approach a limit boulder problem as you might approach a cage fight. Be prepared to squeeze like hell, and expect it to hurt a little bit. Spend rest periods analyzing the previous attempt and considering opportunities for more efficient movement or better

effort. Additional rest, and encouragement from onlookers, can help achieve the high level of arousal required for this activity.

Even more so than for the Limit Bouldering Routine (which is more technically demanding), the training intensity for campusing must be *extremely* high. Summon 100 percent effort and commitment for each attempt, and rest between attempts as much as necessary to allow 100 percent effort. Instead of a drawn-out cage fight, seek a knockout blow with one punch of maximal power. Scream and yell, don't give up, and expect it to hurt when you latch that distant rung. Olympic-style lifters report it can be helpful to get downright angry. Think of something irritating and punish the campus board. Further recommendations on achieving high arousal are provided in **Chapter 15: Bouldering**.

FREQUENCY AND REST

Rest intervals are extremely important for all power training. Elite power athletes in Olympic sports will routinely use work-to-rest ratios on the order of 1:20 for intense power workouts. For a max ladder set that takes about five seconds to complete, a rest period of approximately 100 seconds should be used. There really is no such thing as too much rest, so rest as long as needed to achieve maximum effort for each set.

For these exercises to be effective, the moves must be explosive. Remember that the goal is to complete the plyometric cycle of contractions, meaning that the movements *must* be explosive and you *must* “pop” from rung to rung. Do not latch a rung and hang out for a while before moving to the next. Attempt to bound steadily up the board in one continuous movement. Once fatigue begins to develop, end the workout. At that point, power no longer improves, but injury potential increases significantly, as does recovery time. As a rule of thumb, the beginner routine should include no more than 16 total sets in a single session, the intermediate routine no more than 24, and the advanced routine no more than 32. Perform far fewer sets early in the Power Phase, progressively increasing the number of sets attempted throughout the phase. Finish each workout with two to four supplemental exercises that address specific weaknesses.

“Power training is all about quality of effort so you should take as long as you need to ensure that every attempt is a good one, especially if your aim is to do the problem.” — Ben Moon⁵

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
Su 44	Mo 45	Tu 46	We 47	Th 48	Fr 49	Sa 50
Last Week of Strength Phase						* Warmup Boulder Ladder (WBL) 60m * Campus 15 min
*Skin Care		*Limit Boulder (LB) 90 min *SE	*Skin Care	* WBL 60 min * Campus 20 min * SE	*Skin Care	
*LB Outdoors	*Skin Care	*WBL 60 min *Campus 25 min *SE	*Skin Care		*LB 90 min *SE	*Skin Care

Key: LB - Limit Bouldering m/min - Minutes SE - Supplemental Exercises WBL - Warm-Up Boulder Ladder

Example Power Phase schedule for a novice trainee. Note that campus workouts are followed by two full rest days, whereas Limit Bouldering Routine workouts are followed by a single rest day.

Campus training should be used sparingly because it is extremely high intensity, and has limited specificity to rock climbing compared to other mutually exclusive activities like the Limit Bouldering Routine. Throughout the Power Phase, schedule four to five minutes of bouldering (including Warm-Up Boulder Ladders, Hard Bouldering, and the Limit Bouldering Routine) for every minute of campusing. For example, in one workout, a climber might do 60 minutes of bouldering and 30 minutes of campusing, then during the next workout he might do 90 minutes of bouldering only, for a total of 150 minutes of bouldering and 30 minutes of campusing over the two workouts (a 5-to-1 ratio).

These workouts are stressful to the finger, elbow, and shoulder joints, even though fatigue might not be apparent. Some coaches (in Olympic sports) theorize that since plyometric training elicits significant neurological adaptations, it can be mentally exhausting, resulting in sleepless nights and other signs of mental fatigue. Pay attention to the body's

response to this training and establish rest periods accordingly. If fatigue mounts from session to session, or if explosive power can't be summoned, increase the rest period between workouts. Since a good campus routine should be relatively short by definition, many climbers will be able to get by on only one full rest day (i.e., 46 hours of rest) between workouts. The injury-prone should strongly consider two full rest days (i.e., 70 hours of rest) between workouts. Baseline Power Phases for novice and advanced trainees are shown.

The Power Phase should generally not last more than four weeks. Campus power typically wanes after three to four weeks, while bouldering ability continues to progress as movement skills and core strength improve. Once campus performance plateaus, replace it with the Limit Bouldering Routine, or proceed to the Power-Endurance Phase. The above schedule should vary based on specific climbing goals and on opportunities for outdoor bouldering (three-day weekends), etc.

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
Su 37	Mo 38	Tu 39	We 40	Th 41	Fr 42	Sa 43
Last Week of Strength Phase					* Warmup Boulder Ladder (WBL) 60m * Campus 25m; SE	*Skin Care
	*Limit Boulder (LB) 120 min *SE	*Skin Care	* WBL 60 min * Campus 30 min * SE	*Skin Care		*LB Outdoors
*Skin Care	*WBL 60 min *Campus 35 min *SE	*Skin Care		*WBL 60 min *Campus 40 min *SE	*Skin Care	*LB Outdoors (on boulders or routes)
*Skin Care	*WBL 60 min *Campus 45 min *SE	*Skin Care		*LB 120 min *SE	*Skin Care	PE Phase

Example Power Phase schedule for an advanced trainee.

Max-Recruitment Training with a Hangboard

While not ideal, a hangboard can be used for power training; it's just a matter of varying the number and duration of reps and sets. The major drawback is the lack of dynamic movement, which prevents training the speed element of power. Also, hangboard training is less specific than preferred power-training exercises, such as the Limit Bouldering Routine, that provide a transition between the (hangboard-heavy) Strength Phase and outdoor rock climbing. Continuing on the hangboard during the Power Phase impedes this transition, so if using a hangboard for power training, include other movement-oriented activities as well. That said, the controlled environment that makes the hangboard ideal for strength training can also be leveraged to achieve significant gains in recruitment.

Max-Recruitment Hangboard Routine

This routine is similar to the hangboard routines of the Strength Phase, but with reduced numbers of repetitions and more rest to permit higher intensity efforts. Only extremely well-trained climbers should attempt this workout (those who have used the hangboard for many years or are climbing in the 5.13+ range). As with the finger-strength routines described in **Chapter Six: Strength**, all repetitions are static dead-hangs.

The high intensity (and therefore high amount of supplemental weight) required for this workout can create a significant logistical problem. Well-trained climbers may need to add triple-digit weight loads to achieve the desired intensity when hanging with two hands. For this reason, certain grips should be trained with one hand at a time (normally not advised). If using one-arm hangs, arrange them consecutively within the routine to maintain consistent rest periods for each hand — the routine below provides an example. Also, the higher intensity of this routine increases the importance of a thorough warm-up. Warm-up as per the Advanced Campus Routine, but end the Warm-Up Boulder Ladder one or two V-grades below your flash limit, then begin this workout.

MAX-RECRUITMENT HANGBOARD ROUTINE

WARM-UP:

Phase 1: Low-intensity ARC traversing (15 to 30 Minutes)

- Light, active stretching on the wall
- Progressively increase difficulty over last 10 minutes

Phase 2: Abbreviated Boulder Ladder (20 to 30 Minutes)

- 1 to 2 problems per grade from "V-easy" to one or two grades below flash level
- Increase rest period with difficulty

HANGBOARD ROUTINE:

Exercise	Grip	Set	Hand	Reps	Resistance	Total Set Time	TUT	Rest
1	Warm-up edge	1	Both	5	Baseline	0:45	0:25	3:00
	...	2a	Left	3	½ Baseline +10	0:25	0:15	1:25
	...	2b	Right	3	½ Baseline +10	0:25	0:15	3:00
2	Semi-closed crimp	1a	Left	5	½ Baseline	0:45	0:25	1:05
	...	1b	Right	5	½ Baseline	0:45	0:25	1:15
	...	2a	Left	4	½ Baseline +10	0:35	0:20	1:15
	...	2b	Right	4	½ Baseline +10	0:35	0:20	1:15
	...	3a	Left	3	½ Baseline +20	0:25	0:15	1:25
	...	3b	Right	3	½ Baseline +20	0:25	0:15	3:00
3	Middle-ring	1a	Left	5	½ Baseline	0:45	0:25	1:05
	...	1b	Right	5	½ Baseline	0:45	0:25	1:15
	...	2a	Left	4	½ Baseline +10	0:35	0:20	1:15
	...	2b	Right	4	½ Baseline +10	0:35	0:20	1:15
	...	3a	Left	3	½ Baseline +20	0:25	0:15	1:25
	...	3b	Right	3	½ Baseline +20	0:25	0:15	3:00
4	Closed crimp	1	Both	5	Baseline	0:45	0:25	3:00
	...	2	Both	4	Baseline +15	0:35	0:20	3:00
	...	3	Both	3	Baseline +30	0:25	0:15	3:00
5	Pinch	1	Both	5	Baseline	0:45	0:25	3:00
	...	2	Both	4	Baseline +15	0:35	0:20	3:00
	...	3	Both	3	Baseline +30	0:25	0:15	...

SUPPLEMENTAL EXERCISES:

- 2 to 4 exercises addressing specific physical weaknesses
- 1 to 3 sets each of 8 or fewer repetitions

The Max R Hangboard Routine should consist of a larger warm-up grip, followed by four to five basic grip positions (such as semi-closed crimp, middle-ring two-finger pair, closed crimp, mono, pinch). Complete three sets for each grip, with the first set consisting of five repetitions, the second four reps, and the third set consisting of three reps. Each rep comprises a five-second hang followed by five seconds of rest. Rest three minutes between each set.

Unlike the finger-strength routines described in [Chapter 6: Strength](#), the goal is not to carry cumulative fatigue from one set to the next. The purpose of the first and second sets of each grip position is to prepare the muscles for maximal intensity on the final set. Therefore, the increment of applied load from set to set should be greater than that of the finger-strength routines (for example, if the correct load for a given grip of a finger-strength routine is “body weight” for set one, “body weight +10 pounds” for set two, and “body weight +20 pounds” for set three, the correct load for the Max R routine might be “body weight” for set one, “body weight +15 pounds” for set two, and “body weight +30 pounds” for set three).

SUMMARY

- Power is a primary determinant of climbing ability. Power is the ability to quickly exert high force on a set of holds.
- Not all motor units contribute to every muscle contraction. Recruitment describes the proportion of active motor units contributing to a given muscular contraction. Increasing recruitment results in higher force production, but does little to improve contraction speed, which is critical to latching the target hold during a dynamic movement.
- Contact strength is the amount of force the fingers can exert on a hold during the period of initial contact. The faster such force can be generated, the less force is required to latch the target hold.
- Muscle fibers can be trained to improve certain traits like contraction speed and cross-sectional area.
- Plyometric training is widely regarded as the best method for improving explosive power in all types of athletes because it 1) helps increase muscle-fiber recruitment and 2) helps train muscle fibers to contract more quickly.
- The Limit Bouldering Routine is the most climbing-specific method of training power. Limit boulder problems should be short, featuring one to two dynamic crux moves right at the climber’s physical limit.
- The Campus Board is ideal for climbing-specific plyometric training. Campusing also greatly improves confidence, speed, and accuracy for dynamic movements.
- Focus on performing a few, climbing-specific campus exercises well (with smooth, flowing movement) rather than performing a plethora of exercises poorly.
- Power training must be done when the climber is fresh, and able to move explosively. Once power begins to wane, end the training activity.
- Plan adequate recovery periods between power workouts. Plan additional recovery following each campus session.